



LIMIT and ALIASING

Short Documentation: LIMIT and ALIASING (TechSphere Solutions Database)

This SQL script demonstrates how to use the LIMIT clause and ALIASING in MySQL using the techsphere_solutions database.

The script focuses on:

- Restricting query output
- Sorting and pagination
- Renaming columns and tables for readability
- Combining LIMIT with ORDER BY

1. Database Selection

```
USE techsphere_solutions;
```

What it does:

- Selects the active database
- Ensures all queries run within techsphere_solutions database

2. LIMIT Clause

2.1 Retrieve first 3 employees

```
SELECT *  
FROM employees  
LIMIT 3;
```

What it does:

- Returns only the first 3 rows from the employees table
- Useful when previewing data



3. LIMIT with ORDER BY

3.1 Sort then limit

```
SELECT *  
FROM employees  
ORDER BY first_name  
LIMIT 3;
```

What it does:

- Sorts employees alphabetically
- Returns only the first 3 records after sorting

4. LIMIT with OFFSET

4.1 Skip rows and retrieve specific records

```
SELECT *  
FROM employees  
ORDER BY first_name  
LIMIT 3,2;
```

What it does:

- Skips the first 3 rows
- Returns the next 2 rows

Purpose:

Used for:

- Pagination
- Data browsing
- Page-by-page results



5. Practical LIMIT Example

5.1 Retrieve third earliest hired employee

```
SELECT *  
FROM employees  
ORDER BY hire_date  
LIMIT 3,2;
```

What it does:

- Sorts employees by hire date
- Skips first 2 rows
- Returns the 3rd employee

6. ALIASING

Aliasing temporarily renames:

- Columns
- Tables

This improves readability and cleaner query output.

7. Column Aliasing with AS

7.1 Rename calculated column

```
AVG(base_salary) AS average_salary
```

What it does:

- Renames the output column
- Makes results easier to understand

Instead of:

- AVG(base_salary)

You see:

- average_salary



8. Alias Without AS

```
AVG(base_salary) average_salary
```

What it does:

- Produces same result as using **AS**
- **AS** keyword is optional in MySQL

9. Table Aliasing

9.1 Shortening table names

```
FROM employees e  
JOIN departments d
```

What it does:

- **e** becomes alias for employees
- **d** becomes alias for departments

Benefit:

Makes JOIN queries shorter and easier to read.

10. LIMIT + ORDER BY + ALIAS Together

10.1 Top highest-paying roles

```
ORDER BY salary DESC  
LIMIT 3;
```

What it does:

- Renames columns
- Sorts salaries highest to lowest
- Returns top 3 highest-paying roles



11. Overall, Purpose of the Script

This script demonstrates how SQL is used to:

- Restrict returned rows (LIMIT)
- Sort results (ORDER BY)
- Paginate records (LIMIT offset, count)
- Rename columns and tables (ALIASING)
- Improve readability of SQL queries

12. Key Learning Outcomes

After running this script, you understand:

- How to limit query output
- How pagination works in SQL
- How aliasing improves readability
- How sorting and limiting work together
- How professional SQL queries are structured

13. Summary Insight

LIMIT controls how many rows are returned, while aliasing improves query readability by assigning temporary names to columns and tables. Together, they help create cleaner and more efficient SQL queries.